Approaching A Theory of Emotion:

An Interview With Candace Pert, Ph.D

By Lynn Grodzki, Fellow

The New Identify Process (NIP) and other forms of emotive psychotherapy embrace the healing tradition of *catharsis*--intense emotional expression is elicited within a contained therapeutic environment. This emotive therapy follows in a direct line from the earliest forms of ancient healing arts through recent scientific studies exploring the link between body and mind. The challenge for clinicians in answering the criticism about the use of catharsis is to conceptually bridge past and present in evaluating emotive methods.

Happily, an unexpected voice has joined the debate about the importance of emotional therapy. Candace Pert, Ph.D., researcher and pharmacologist, may help point the way to a resolution of a problem that has faced clinicians using emotive methods for the last 100 years. In adding her biochemical research perspective to the discussion of the meaning and value of catharsis, she is addressing a weak point and the biggest problem that exists in the field of emotive, experiential psychotherapy. She is helping to formulate, for the first time, a unified theory of emotion.

First, a bit of history. Although the use of catharsis was a key element of treatment during the first 200 years of early psychotherapy (with Mesmer, Charcot, Janet, and Bruer), Freud's rejection of this cathartic method within psychoanalysis and his reliance on free association, "The talking cure" as a sufficient form of abreaction, spread until dominating the field. By 1920, methods of emotive psychotherapy moved to the fringes of conventional psychological practice. Freud gave as one of his reasons for rejecting emotive methods his frustration as a neurologist in trying to theorize about the workings of emotion. Although some of his colleagues continued to rely on methods of catharsis (notably Ferenczi, Brown and Reich) and although a second wave of interest sparked the development of additional methods in the early 1950's (by Janov, Lowen, Perls, Casriel and Jackins) the academic literature continues to reject catharsis, following Freud. Methods of emotive psychotherapy, when mentioned, are usually discounted as unproved and ineffective at best, or counterproductive and harmful at worst. Currently, the criticism of emotive therapy is based on the results of often flawed, past research about catharsis.

In some studies, catharsis is misdefined to mean any kind of ventilation (from watching a wrestling match to screaming, to hitting another person). Because clients require a safe space (environmental containment) in order to achieve a true experience of catharsis, the results that clinicians can produce in their office settings are hard to reproduce in laboratory settings. But the biggest hurdle to researching and validating emotional methods has been the vagueness about emotion itself. Until recently, little has been understood from a scientific basis about what emotion is and is not.

Psychological textbooks published only thirty years ago state, "Emotion is virtually impossible to define . . . except in terms of conflicting theories" and "No genuine order can be discerned within the field." As long as emotion remains an abstraction, lacking a unified theory base, it is impossible to research and validate methods of emotive therapy. The kinds of questions that need to be answered include: how emotion is manifest, how memory and emotion interact, whether emotion is concrete (real) or conceptual (a construct), if concrete, how emotion acts in the body, and how unexpressed emotion is stored.

Enter into this discussion Dr Candace Pert. For the past twenty years, Pert has been studying the movement of amino acid chains in the human body. In the process, she is unraveling the mystery of mind-body communication and changing forever the way we understand emotion.

For Pert, pharmacologist and professor at Georgetown University, the mind is not just in the brain -- it is also in the body. The vehicle that the mind and body use to communicate with each other is the chemistry of emotion. The chemicals in question are molecules, short chains of amino acids called peptides and receptors, that she believes to be the "biochemical correlate of emotion." The peptides can be found in your brain, but also in your stomach, your muscles, your glands and all your major organs, sending messages back and forth. After decades of research, Pert is finally able to make clear how emotion creates the bridge between mind and body.

Candace Pert lives in the world where emotions make scientific sense. As former Chief of Brain Biochemistry at the NIH for 13 years, she studied the inner workings of the body with an eye towards identifying and locating peptides and receptors. She became convinced these chemicals were the physical manifestation of emotion. In 1993, Pert appeared on Bill Moyer's landmark TV program *Healing and Mind*, where she explained her theories of emotion to a national audience. She attracted attention for being that rare scientist who can explain their work to a lay audience with a sense of humor and passion. These days Pert spends substantial amount of time in Rockville, Maryland, as a consultant on the trials of a new drug, Peptide T, that is part of a non-toxic AIDS therapy. She takes some time from her research and teaching schedule to lecture internationally on the issues of neuropeptides and mind-body communications.

I began to correspond with Pert several years ago, and in May of 1995, as a result of her desire to be part of the 1995 ISNIP Conference, we sat down to talk about a subject that interests both of us: the need for a unified theory of emotion. She offered some new, startling insights of her own that explain how experiential forms of psychotherapy and alternative medicine work. What follows is a portion of our discussion.

Lynn Grodzki: How do you understand the connection between memory and emotion?

Candace Pert: Experiments show that the hippocampus area of the brain [part of the limbic system] is the access or gateway into the whole emotional experience. Almost every variety of peptide receptor is found in the hippocampus. Through the peptide network, which is anything that has peptide receptors on it, you can access different memories, mood states or

developmental stages. Strong emotions are the key variable that make us bother to remember things.

There is a lot of evidence that memory occurs at the point of synapse, there are changes that take place in the receptors. The sensitivity of the receptors are part of memory and pattern storage. But the peptide network expends beyond the hippocampus, to organs, tissue, skin, muscle and endocrine glands. They all have peptides receptors on them and can access and store emotional information. This means this emotional memory is stored in many places in the body, not just the brain. The autonomic nervous system is pivotal to this entire understanding. Its importance is much more subtle than has been thought. Every peptide that I have every mapped and more can be found in the autonomic nervous system. There is an emotional coding to the way our autonomic patterns are elaborated.

LG: The autonomic nervous system includes the spinal cord and the ganglion that are down either side. Is it possible that emotion could be stored in places like this indefinitely?

CP: Absolutely. Emotional memories are our earliest memories. One of my earliest memories is that I struck a match when my mother was making dinner. I just started a tiny fire, and she came over and put it out with her dishrag. I can still see the terror in her face. I think I must have been one year old. Emotional memories are long term memories, stored where we need them, for survival.

LG: Let's say you had forgotten this memory and you are in a situation where something similar happens, perhaps your own daughter plays with matches and you find your reaction has an intensity that suggests an earlier incident was attached to it. How is early emotional memory retrieved in the body?

CP: You can access emotional memory anywhere in the peptide/receptor network, in any number of ways. For example, if you have a memory that has to do with food and eating, you might access it by the nerves hooked up to the pancreas. You can access through any nodal point in the neural loop. Nodal points are places where there is a lot of convergent information with many different peptide receptors. In these nodal points there is potential for emotional regulation and conditioning.

LG: So we are programmed to be able to repeat emotional experience and we can access it through the body in many ways. What happens to emotions that are not able to be fully expressed?

CP: I have a whole theory about this. I believe that emotion is not fully expressed until it reaches consciousness. When I speak of consciousness, I include the entire body. I believe that unexpressed emotion is in process of traveling up the neural access. By traveling, I mean coming from the periphery, up the spinal cord, up into the brain. When emotion moves up, it can be expressed. It takes a certain amount of energy from our bodies to keep the emotion unexpressed. There are inhibitory chemicals and impulses that function to keep the emotion and information down. I think unexpressed emotions are literally lodged lower in the body.

In my mind, there are levels of integration. You are integrating lower brain areas when you

move the emotion up and get it into consciousness. That's where you begin comprehension. I often tell a story in my lectures. I show a picture of a woman with hot coffee, who has dropped the cup and burned herself. She reacts to the scalding coffee by being startled and feeling pain. The emotional reflex moves up and up and up the body. When it finally gets to the level of the thalamus she says, "Oh, it's hotter than it usually is." But then I make a joke. I say, "It's only when it gets all the way up to the cortex that she can actually blame her husband." That's where we put the whole spin on it. Unexpressed emotions are buried in the body -- way, deep down in the circuitry of the organs, or the GI tract, or a loop in a ganglium. We even know what the memory storage looks like. It's protein molecules coupled up to receptors. Some thought it only gets stored in the brain. But it looks like that in the body, too. Your memories can get stored that way in a pancreas, for example.

LG: There is a belief that unexpressed emotion is harmful to the mind and body. IF you haven't fully grieved a loss, for example, your weakened immune system might make you a candidate for an illness, like cancer. How do you understand it, as a scientist?

CP: It think there is overwhelming evidence that unexpressed emotion causes illness. I'm a molecular Reichian!

LG: Reich had a model of working with emotion that is sometimes called the "conflict model" of catharsis. He thought there were two psychic forces at work in every individual. One is the force that wants to express emotion. The other is the force that seeks to prevent its expression, which he termed resistance. He thought the pressure of the two forces caused stasis, so his therapy techniques were designed to exhaust and weaken the resistance, to allow emotional expression to occur.

CP: I see it this way. The raw emotion is working to be expressed in the body. It's always moving up the neural access. Up the chakras, if you will, but really up the spinal chord. The need to resist it is coming from the cortex. All the brain; rationalizations are pushing the energy down. The cortex resistance is an attempt to prevent overload. It's stingy about what information is allowed up into the cortex. It's always a struggle in the body. The real, true emotions that need to be expressed are in the body, trying to move up and be expressed and thereby integrated. That's why I believe psychoanalysis in a vacuum doesn't work. You are spending all your time in your cortex, rather than in your body. You are adding to the resistance.

LG: You suggest a vertical model of catharsis, letting the emotion move up the body, perhaps finding ways to relax the cortex to allow the unexpressed emotion to be first experienced and then cognitively integrated.

CP: Let the emotion all bubble up. Let the chips fall where they may. My personal experience using catharsis was with the New Identify Process. I think the NIP bonding might serve to relax the cortex and let the emotion come through. I believe that the process of catharsis is not complete without saying things, because we must involve speech and the cortex, to know that the emotion has come all the way up and is being processed at the highest level. To feel and understand means you have worked it all the way through. It's bubbled all the way to the surface. You're integrating at higher and higher levels in the body, bringing emotion into

consciousness.

Candace Pert, Ph.D. was the keynote speaker at the 1995 ISNIP Conference, September 23, 1995

For more information about the NIP, or to become a member of the American Chapter and receive the quarterly newsletter, please contact Yetta Lautenschlager, President of ASNIP at toll free phone number 1-888-912-1891. You may also contact Lynn Grodzki, LCSW, Tel. (301) 434-0766, or e-mail her at Lgrodzki@erols.com